Computer Science & IT



Database Management System

File organization & Indexing

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- Topic: Database, File and Records
- Topic: Types of Records
- Topic: Blocking Factor
- Topic: Organization of Records
- Topic: Search-key & Type of File

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Topics to be Covered



- Topic: IO cost without index file
- Topic: Index file
- Topic: IO cost with indexing
- Topic: Categories of index
- Topic: Types of index

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Topic : IO cost without indexing

Let 'N' is the number of secondary memory blocks required to store a file.

(i) Worst Case IO Cost = N

(w.r.t. Unordered file)

∴ Linear Search

{ in worst case we may have to transfer all the blocks of the file from secondary memory to main memory in order to search for a record }.

(ii) Worst Case IO Cost = $\lceil \log_2 N \rceil$

(w.r.t. Ordered file)

∴ Binary Search is possible

Ordered file

Advantages: Binary search is possible, i.e. less IO cost

Disadvantage: Insertion of new record will be a costly oph.

Unordered file (Heap file)

Advantage: Insertion will always be performed in the last block of the file, i.e. Insertion will be easy.

Disadvantage: More IO cost.



Topic : Index file

* Index file can be used to reduce the IO Cost.
* Each entry of the index file will consist of two fields (i) Search key field (ii) Pointer to the record/block corresponding to the search-key.



Topic : Index file

Search key	Pointer w.r.t. search key

Index file { let one index file entry for each DB record }

Let,
Blocking factor w.r.t Index file = 6

Search key

Eid		
9		
13		
18		
1		
15		
10		
4		
16		
2		
14		
5		
3		
17		
7		
11		
6		
12		
8		

Database file

B_1 B_2 B_3 B_4 B_5 B_6

There may be various other attributes apart from search key in a database record

∴ Size of each record = Summation of size of each attribute/field

3 records in each block of DB file

∴ Blocking factor = 3 { w.r.t database file }



Topic : Index file

(i) Index file entry size $=$ (Search key value size) + (Pointer size)
(i.e. size of each entry in index file)

(ii) There are various fields in each record of database file
$\therefore$ (Index file entry size) $<$ (DB file record size)

(iii) Same disk blocks are used to store both database records as well as index file entries.
$\therefore$ Index file block size = DB file block size

(iv) No of entries in the index file $\leq$ No. of records in DB file

It will be '=' if index is dense index

It will be '<' if index is sparse index.



Topic : Index file

Blocking factor {w.r.t. database file} = $\frac{\text{Block size}}{\text{record size}}$	Blocking factor {w.r.t. index file} = $\frac{\text{Block size}}{\text{Index file entry size}}$ ↓ i.e. No. of entries in each block

(V) Blocking factor of index file $\left(\frac{\text{Block size}}{\text{Index file entry size}}\right)$ > Blocking factor of DB file $\left(\frac{\text{Block size}}{\text{Record size}}\right)$



Topic : Index file

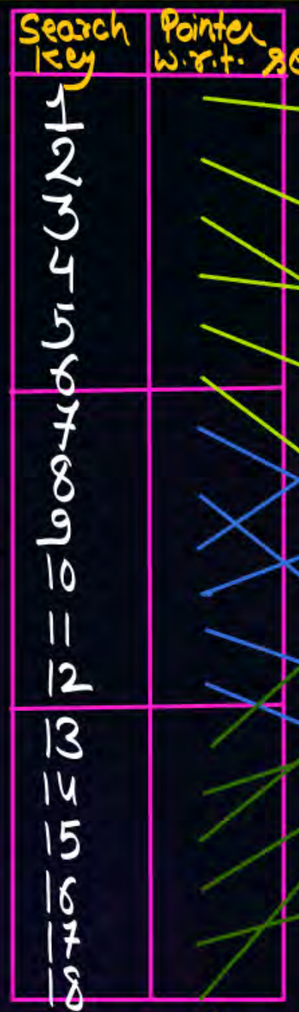
Number of blocks required to store Complete DB file	$= \frac{\text{Total No. of DB records}}{\text{Blocking factor of DB file}}$	↑ ↓
Number of blocks required to store Index file	$= \frac{\text{No. of index file entries}}{\text{Blocking Factor of index file}}$	↓ ↑

(vi) Number of Index blocks required to store the index file >> Number of blocks required to store DB file.

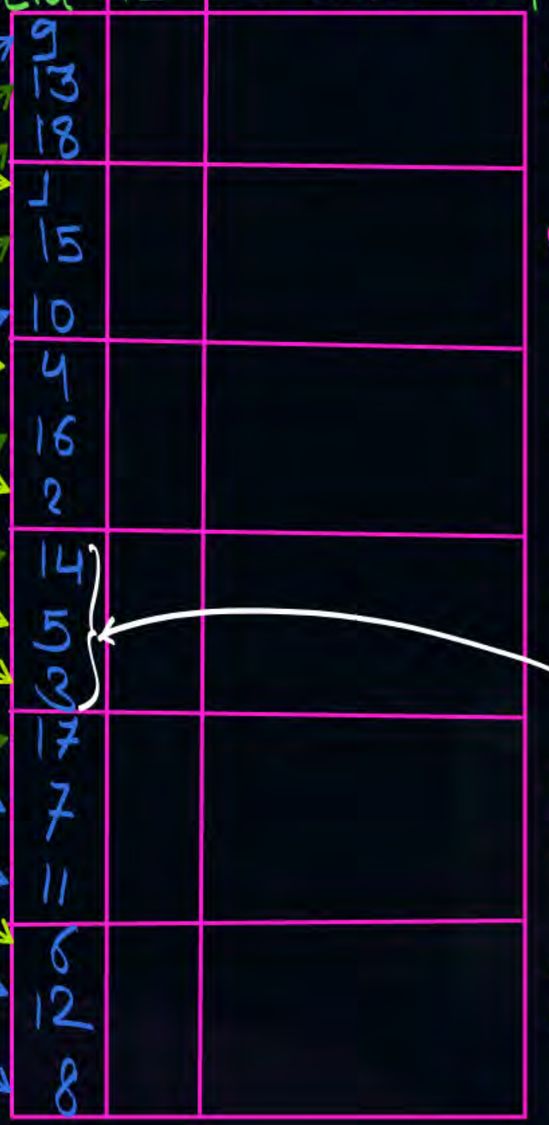
(vii) In general, Index file will be Ordered based on search key



Topic : Index file



Search key



There may be various other attributes apart from search key in a database record

∴ Size of each record = Summation of size of each attribute/field

Let,
Blocking factor w.r.t Index file = 6

3 records in each block of DB file

∴ Blocking factor = 3 {w.r.t database file}

Index file {let one index file entry for each DB record}



Topic : IO cost with indexing

* Let 'N' be the number of disk block required to store the DB file
* And let 'M' is the number of disk blocks required to store the index file,

We know, $M < N$. and index file will be ordered based on search key.

∴ Worst Case IO Cost (with index file) $= \lceil \log_2 M \rceil + 1$

($\lceil \log_2 M \rceil$: No. of blocks in index file)

(1 : DB block access w.r.t address provided by the pointer of Index file)



Topic : Categories of Index

There are two categories of Index:.

(1) Dense Index { for each record of the database file there will be one entry in the index file }

(2) Sparse Index { for a collection of records in the DB file there will be one entry in the index file }



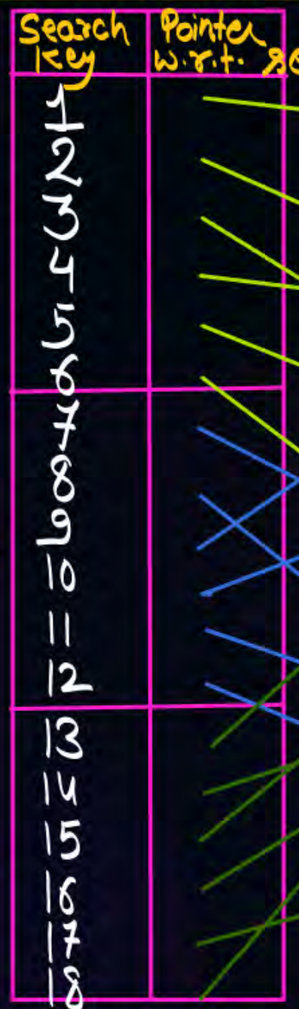
Topic : Dense Index

* In dense index for each record of the DB file there will be one entry in index file.

∴ No. of entries in the index file = No. of records in the database file



Topic : Example of Dense Index



Let,
Blocking factor w.r.t Index file = 6

Index file (Dense Index) {let one index file entry for each DB record}



There may be various other attributes apart from search key in a database record

∴ Size of each record = Summation of size of each attribute/field

3 records in each block of DB file

∴ Blocking factor = 3 {w.r.t database file}



Topic : Sparse Index

* For a Collection of database records there will be One entry in the index file.

∴ No. of entries in index file < No. of records in DB file

By default one entry in the index file for each block of the DB file

i.e. No. of entries in the index file = No. of blocks in the DB file



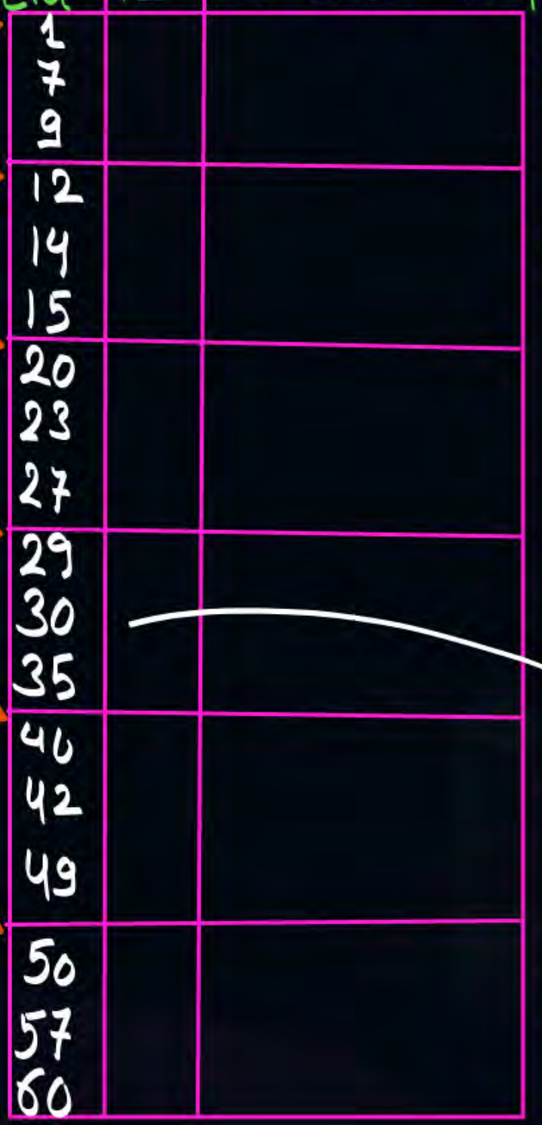
Topic : Example of Sparse index



Let,
Blocking factor w.r.t Index file = 6

Sparse index file
{1-entry per database block}

{No. of blocks required in the index file w.r.t Sparse index} ≤ {No. of blocks required for the index file using dense index}



There may be various other attributes apart from search key in a database record

∴ Size of each record = Summation of size of each attribute/field

3 records in each block of DB file

∴ Blocking factor = 3 {w.r.t database file}

Q: Consider

Disk block size = 1000 Byte

Database record size = 100 Byte

Let, key value size = 12 Byte

& Pointer size = 8 Byte

and No. of records in the DB file = 10,000

(i) How many disk blocks are required for a dense index file

(ii) How many disk blocks are required for a sparse index file

(i) No. of Blocks required for dense index file

$$= \frac{\text{No. of entries in the dense index file}}{\text{Blocking factor of index block}}$$

$$= \frac{\text{No. of records in the DB file}}{\left\{ \frac{\text{Block size}}{\text{index file entry size}} \right\}}$$

$$= \frac{10,000}{\left\{ \frac{1000}{\text{key value size} + \text{Pointer size}} \right\}} = \frac{10,000}{\left\{ \frac{1000}{12+8} \right\}} = \frac{10000}{50} = 200$$

(ii) No. of Blocks required for Sparse index file

$$= \frac{\text{No. of entries in the Sparse : index file}}{\text{Blocking Factor of index block}}$$

$$= \frac{\text{No. of blocks required to store DB file}}{\left\{\frac{\text{Block size}}{\text{index file entry size}}\right\}}$$

$$= \frac{\left(\frac{\text{No. of records in DB file}}{\text{Blocking Factor of DB block}}\right)}{\left\{\frac{1000}{\text{key}_{\text{size}} \text{ value} + \text{Pointer size}}\right\}} = \frac{\left(\frac{10,000}{10}\right)}{\left(\frac{1000}{12+8}\right)} = \frac{1000}{50}$$

$$= \boxed{20}$$

$$= \frac{\text{Block size}}{\text{Record size}}$$

$$= \frac{1000 \text{ Byte}}{100 \text{ Byte}} = (10)$$



Topic : Types of Index

- Types of Index
  - Single level Index
    - Primary Index
    - Clustering Index
    - Secondary index
  - Multi-level Index
    - Static Multi-level Index
    - Dynamic Multi-level index
      - B tree index
      - B^+ tree index



2 mins Summary

- **Topic** IO cost without index file
- **Topic** Index file
- **Topic** IO cost with indexing
- **Topic** Categories of index
- **Topic** Types of index

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THANK - YOU

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